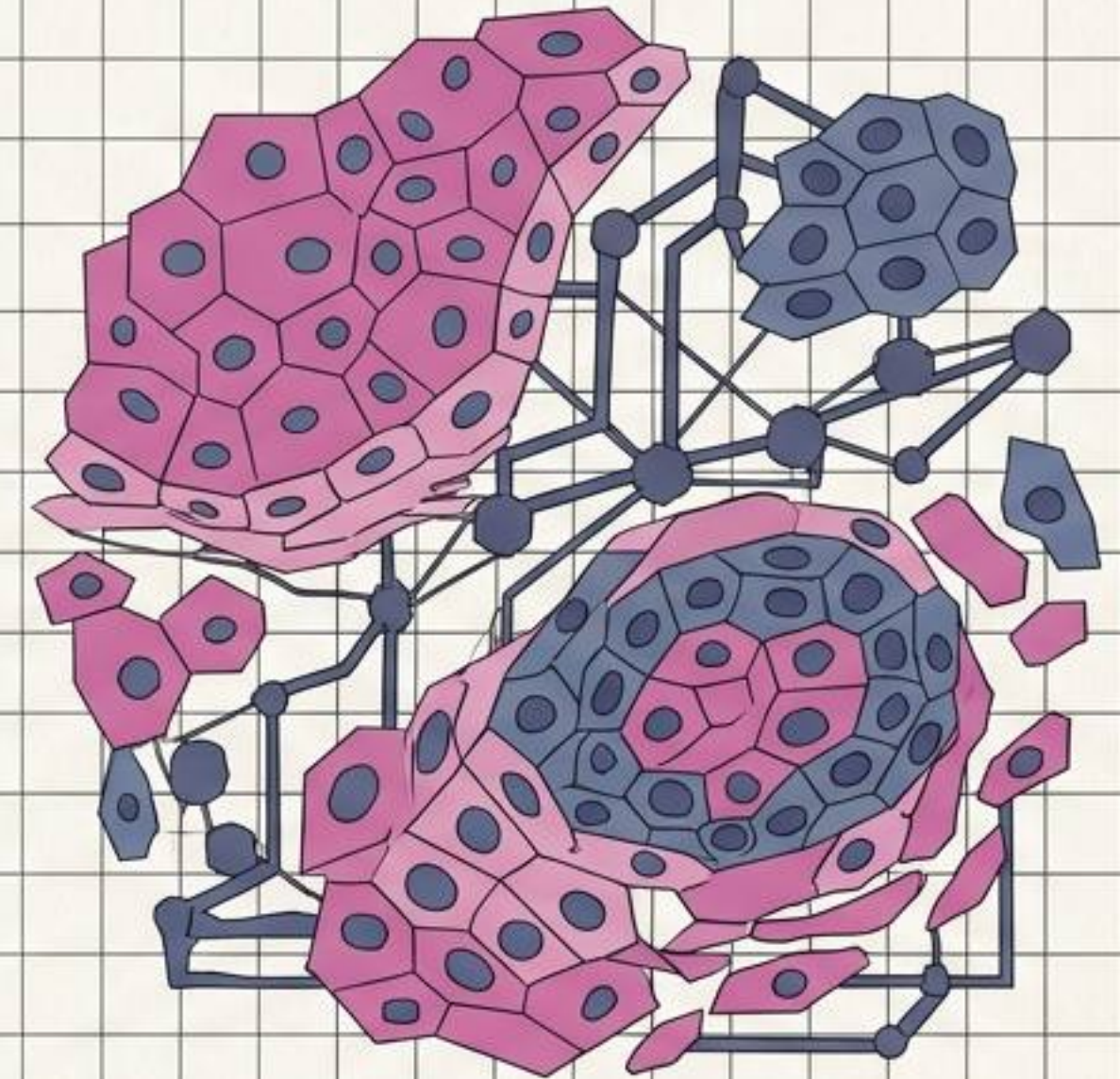


Improving Underrepresented and Difficult Classes in Computational Pathology

Progress Update | Yannik Queisler | 2026-06-30

Thesis Statement: Investigating why specific classes underperform on CPath datasets and establishing how to fix them using frozen foundation-model (Virchow2) features.



1. Isolate
underperforming
factors

2. Address
independently

3. Formulate
Joint Mitigation
Method

Defining the Generalization Imperative

High-Level Goal: Improve performance on underrepresented and difficult classes by addressing all interacting factors simultaneously.

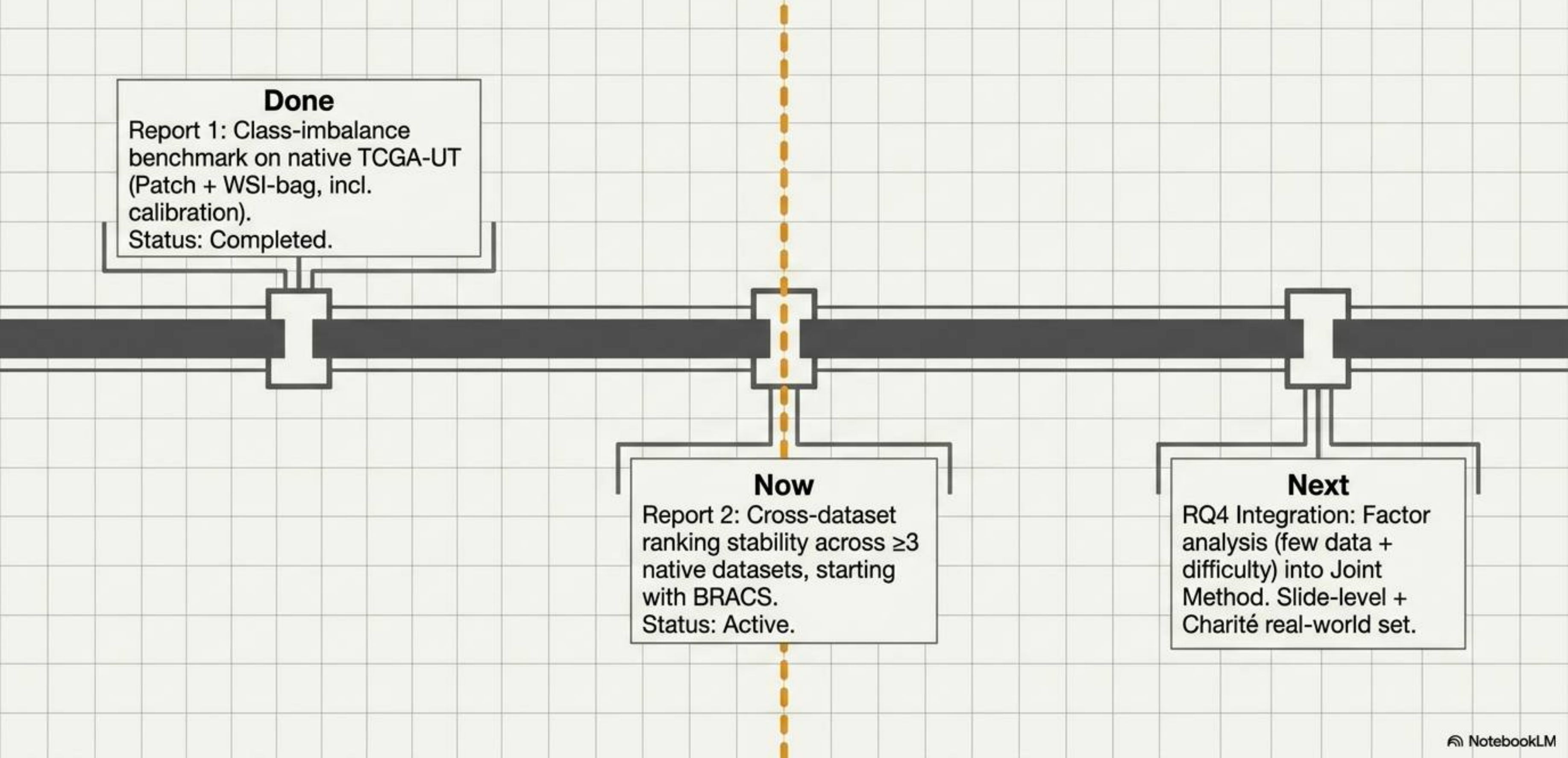
RQ4 — Joint Method: A single architecture addressing all factors simultaneously while maintaining strict calibration.

RQ2 — Independent Mitigation: Efficacy of isolated fixes & cross-dataset stability.

RQ1 — Causes:
Class Imbalance | Limited Data |
Class Difficulty
(Label noise? Patch vs WSI-bag?)

RQ3 — Interaction:
Disentangling of factors.

Project Trajectory & Current Position



Report 1 Setup: Divergent Regimes, Unified Baseline

Virchow2 (Frozen)

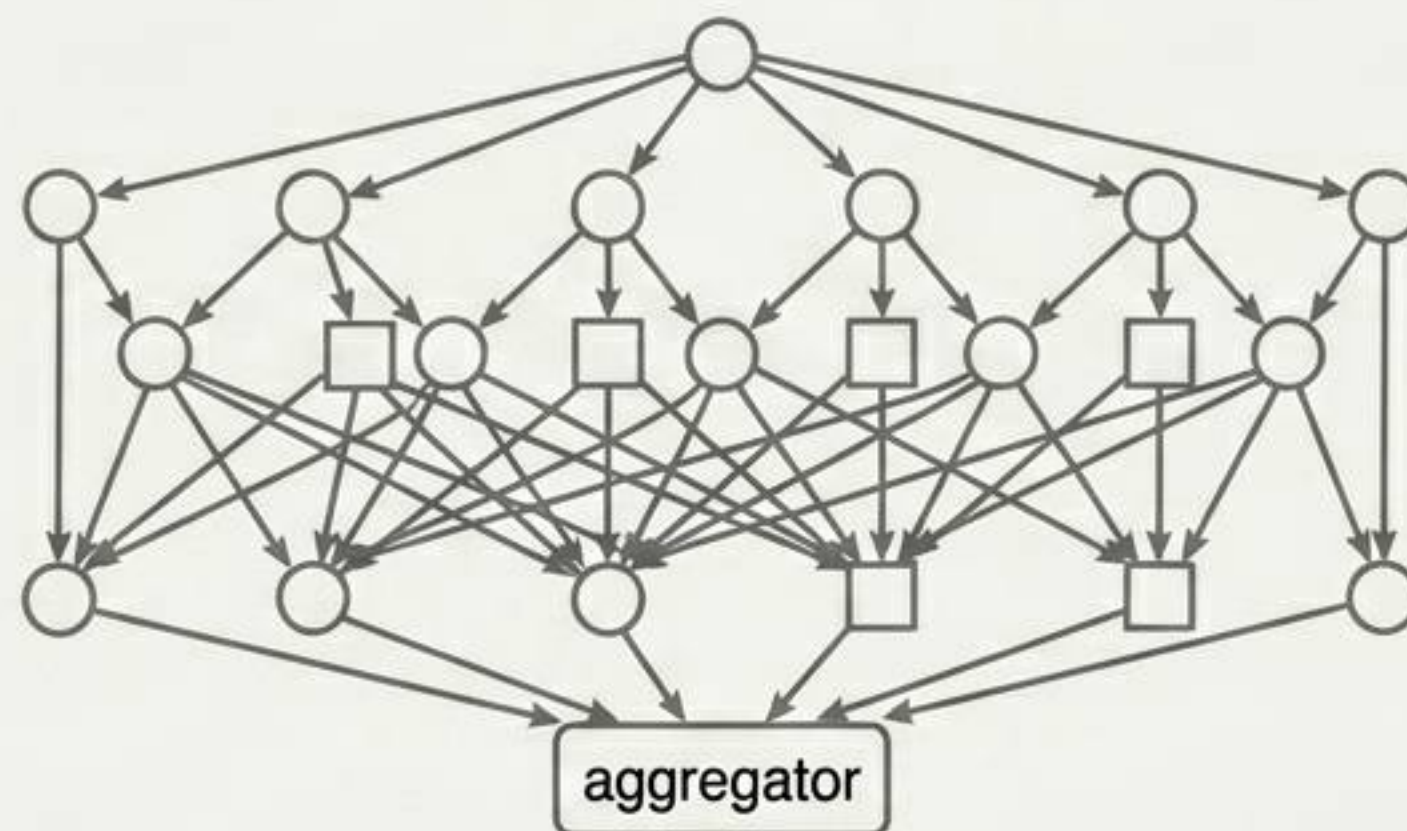
TCGA-UT | 32 Cancer Types | Native $\approx 28:1$ Imbalance | Case-disjoint, 3 seeds, ≤ 30 instances/slide.

Patch Regime



Narrow band architecture.
Directly classifies individual tiles.

WSI-Bag Regime



Multi-instance aggregation.
Structurally yields ~ 10 BAcc points higher
than the patch regime.

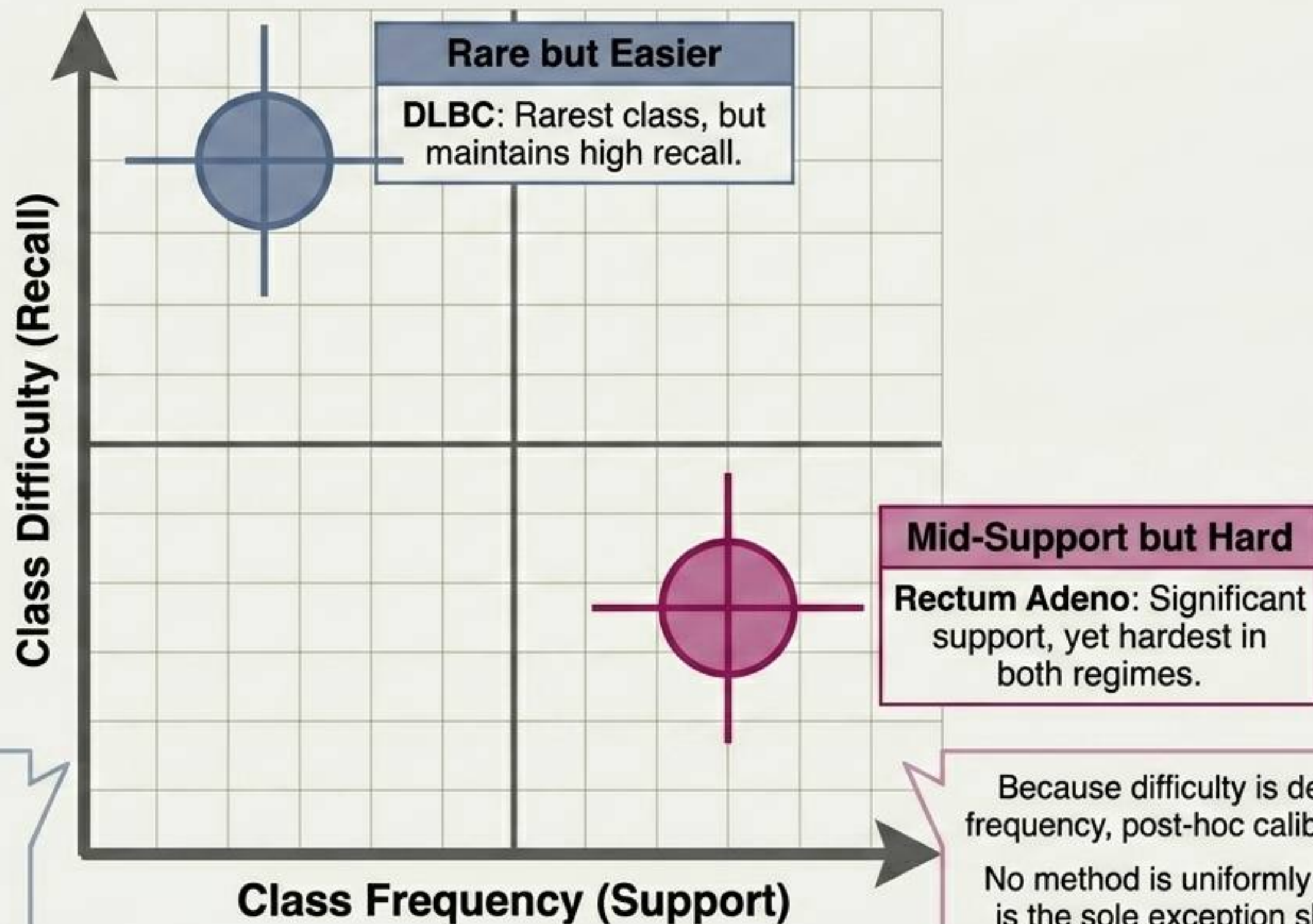
These two distinct regimes have never been directly compared on this frozen-encoder baseline.

Mechanism Matrix: Head Optimization Outperforms Data Manipulation

Patch Regime	WSI-Bag Regime
Best Macro F1: CFAL	Best Mean BAcc & Stability: RankMix
Best BAcc & Calibration (Raw NLL/Brier): OKO	
Note: Resampling, reweighting, focal, ProGAN functionally $\approx$ Cross-Entropy.	Best Mean Macro F1: MDE-MIL (gaps small vs seed spread).

With a frozen encoder, performance gains stem from the head/objective. Because the best mechanism differs fundamentally by regime, native input evaluation is mandatory.

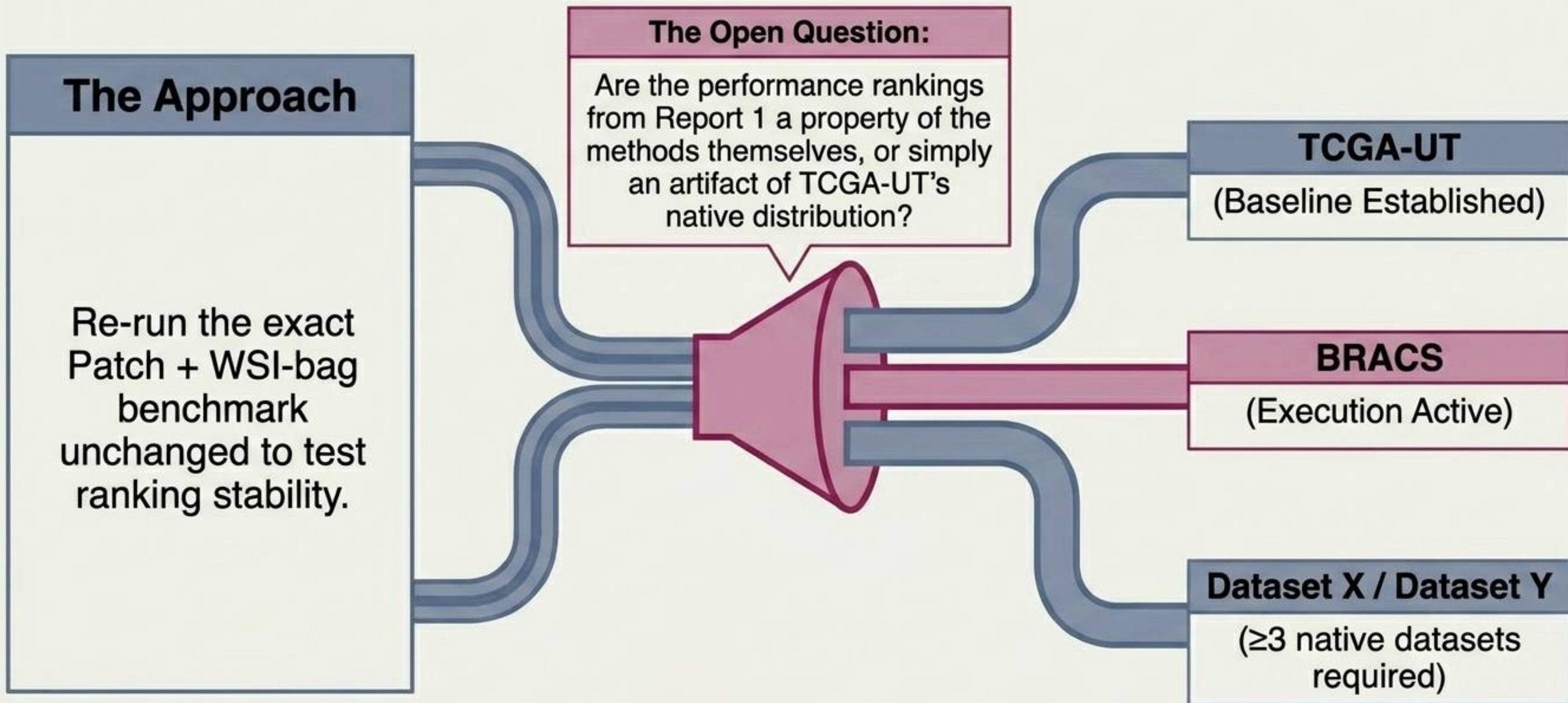
The Calibration Imperative: Difficulty $\neq$ Rarity



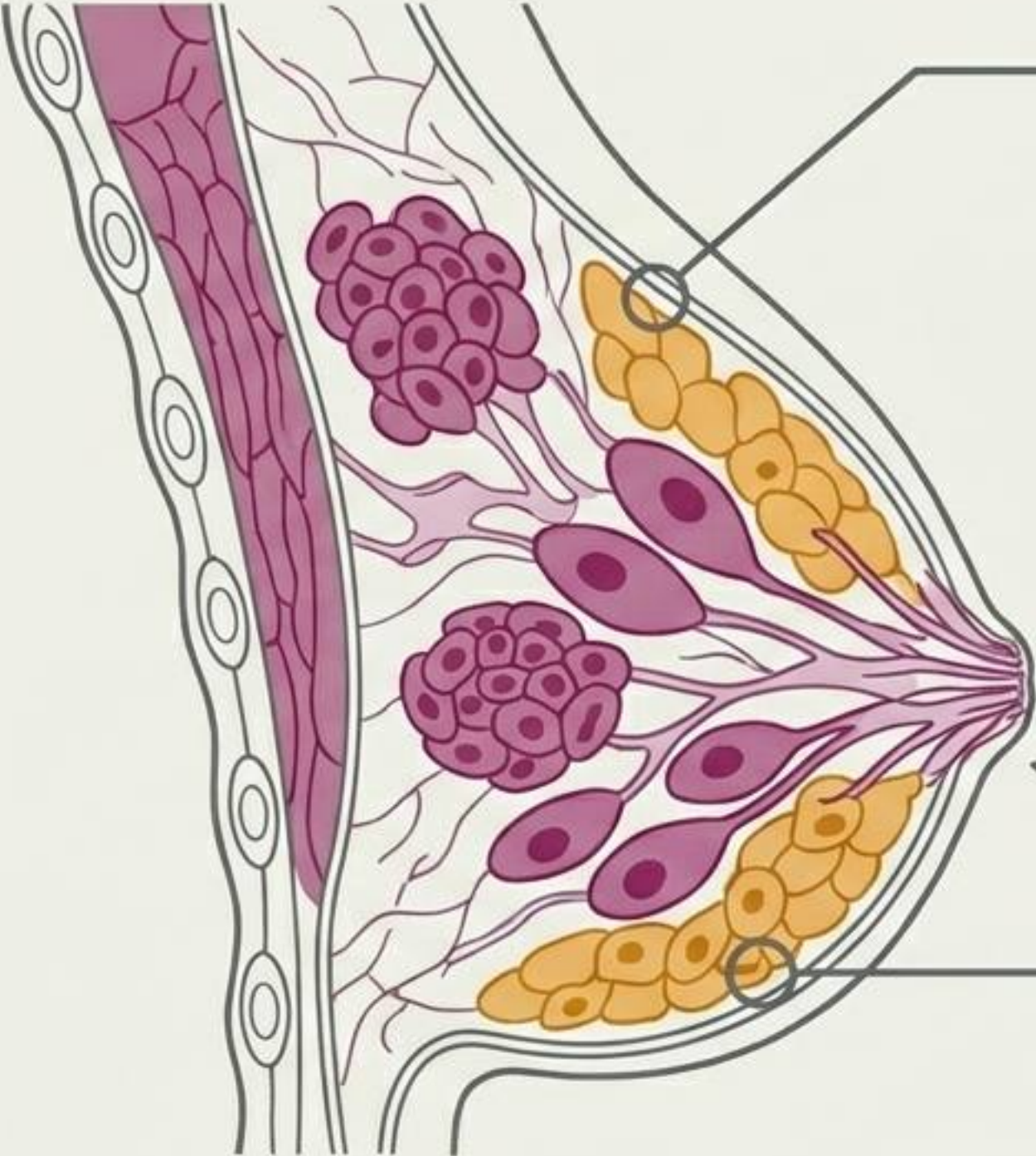
Statistical Reality:
 $\rho \approx 0.30-0.33$
(Weak correlation between support and recall).

Because difficulty is decoupled from frequency, post-hoc calibration is critical. No method is uniformly best, but OKO is the sole exception showing strong discrimination and likelihood. CFAL, while strong in F1, remains overconfident.

Pivot to Report 2: The Generalization Test



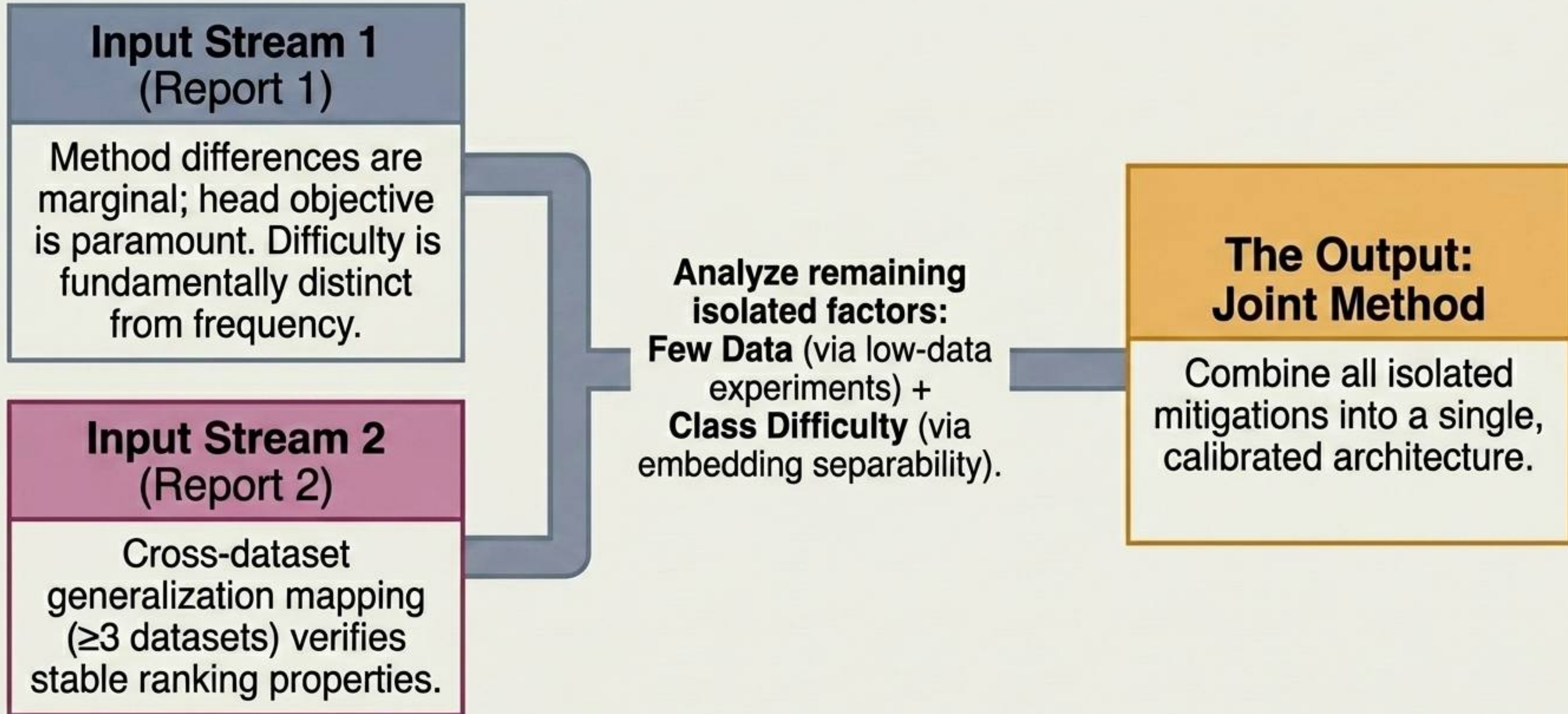
Step One: The BRACS Challenge



Dataset Profile:	Breast-lesion H&E
	547 WSIs
	4539 ROIs
	189 Patients
	7 Subtypes
Clinical Justification Atypical lesions (FEA, ADH) are highly scarce but clinically critical, providing a highly meaningful native imbalance for testing.	
Constraints: Native distribution retained, ROI-tiled (≤ 30 tiles/ROI), patient-disjoint, frozen Virchow2.	

[OK] BRACS staging fixed (commits pushed) | [RUNNING] Benchmark queued; results pending.

Synthesis & Trajectory to RQ4



Strategic Alignment & Advisor Discussion

1.

The Generalization Map

What are the best candidate datasets to fulfill the remaining slots for the cross-dataset benchmark and controlled profiles?

2.

The Factor Audit

Reviewing our nodes (Imbalance, Data Scarcity, Difficulty)—is the performance-degrading factor list complete?

3.

Real-World Deployment

What is the current status and availability of the contributed datasets and the Charité real-world set?